

Improved Trail-Based Selector Study

Booking (Ahead of time)

- Send consent form

Checklist (Preparation Before Study)

- A. 3 Copies Consent Form
- B. Charge any devices needed
- C. 2 Pens
- D. Verify with each of the X participants:
 - Time and place of meeting
 - Contact info:
 - Phone number
 - Address
 - Who will be there
 - Go over any consent questions if lingering
- E. Surveys:
 - 2 background questionnaires

Setup

- Set up design materials
 - Assign the participant an anonymous ID
-

Consent (~10min)

Introduction:

Welcome! Thank you for participating in this study. Throughout the study, I may need to read from a script to ensure that I keep our session on track.

The purpose of this study is to explore how users interact with cross-based / trail-based selection techniques compared to traditional point-and-click methods. By analyzing speed, accuracy, and user confidence, we aim to refine this selection approach, making it more intuitive, efficient, and user-friendly for respective real-world applications.

This study is being conducted by myself, George Morales.

Important Information

*This study involves answering a few questions about your background and experience with pointing devices and interface navigation. You will then complete a series of target selection tasks using a **crossing-based selection method**, where you will select targets by dragging over them rather than clicking directly.*

*We will collect data on your **speed, accuracy, and selection patterns** to better understand performance and usability.*

Your participation is entirely voluntary, and you may choose to withdraw at any time. If you have any questions before we begin, feel free to ask.

Study Procedure

*Today's session consists of two activities. The first will be a quick demonstration of the use of your favorite burhs. The latter will be focused on evaluating a **trail-based selection technique** for pointing interactions.*

*You will begin with the **training phase** to familiarize yourself with the selection method. In this phase, you will practice using a **selection brush** that allows you to select targets by **pressing CTRL once, pressing the left mouse button, and dragging over the highlighted rectangles only**. You can also **right-click when in brush-mode** to deselect any mistakenly selected items.*

*After training, you will be able to proceed to the final **testing phase**, where you will complete a series of trials with randomly placed targets. Your goal is to **select the required targets while avoiding unnecessary selections**, as quickly and accurately as possible. Each trial will have a **randomized arrangement of targets**, and you will be measured on:*

- **Selection speed** – How quickly you complete the task.
- **Accuracy** – How many required targets you select correctly.
- **Errors** – How many unnecessary selections were made.
- **Deselection patterns** – How often you undo selections.

*Once all trials are completed, you will receive a **summary of your results**, showing your performance metrics, including accuracy rates and time taken.*

*At the end of the session, you will be asked to provide **feedback on your experience**, share any **challenges** you encountered, and suggest potential **improvements**.*

Design Presentation (~5-10min.)

This session will be conducted on my laptop. The experiment has one part: a target selection task.

They will be asked to perform a series of target selection tasks, selecting 'highlighted' rectangles from a group of both non-highlighted (green) and highlighted (yellow) targets.

The session includes a practice round followed by 10 test trials. Data on reaction time, accuracy, and movement patterns will be recorded. A summary at the end will show performance results.

Notes on User's Favorite Brush

I'd like to ask you a few questions about your favorite drawing tool and in fact if you could demo it to me on my device if possible.

- 1) What specifically do you like about this tool?
 - 2) What do you wish this tool could do right now that it cannot do?
-

User Feedback on my Trail-Based Selector

I'd like to proceed with my brush demo. While you play around with the tool, I will be asking questions and taking notes.

At the end of the trial, I will ask you the following questions:

- 1) What did you like about my brush?
- 2) What did you not like about my brush?
- 3) What could be improved in terms of functionality or ease of use?